

Name	Subject Class	Class	Index Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2013

BIOLOGY

9648/03

Applications Paper and Planning Question

3 Sept 2013
2 hours

PAPER 3

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
TOTAL	72

This document consists of **16** printed pages.

1. Embryonic stem (ES) cells are highly regarded in research because their pluripotency offers the potential for a wide range of therapeutic and research applications. However, ethical concerns have been raised with regards to the source of ES cells.

Scientists have come up with an alternative method of generating pluripotent cells, which is to genetically reprogramme adult cells to an embryonic stem cell-like state. Genetic reprogramming is carried out by using deactivated retroviruses to introduce the genes of four transcription factors into adult cells from a patient (Fig. 1.1). The reprogrammed cells, called induced pluripotent stem (iPS) cells, are specific to the patient from which the adult cells were taken.

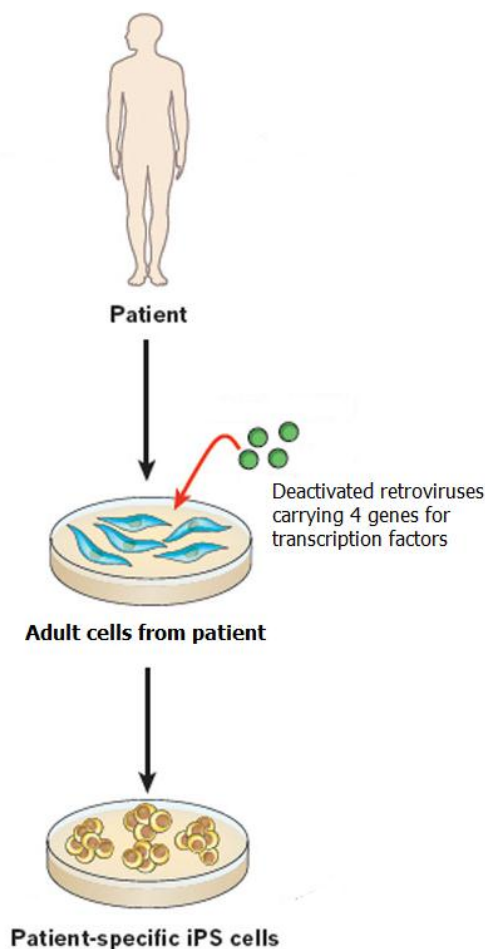


Fig. 1.1

- (a) (i) Describe what is meant by the term 'pluripotent'.

[1]

- (ii) Suggest **one** reason why the use of iPS cells evades the ethical concerns regarding the source of ES cells.

[1]

- (iii) Give **three** reasons why deactivated retroviruses are used for genetic reprogramming of the adult cells rather than active retroviruses or liposomes.

[3]

- (iv) Suggest how the expression of such a small number of transcription factors in adult cells could genetically reprogramme these adult cells to an embryonic stem cell-like state.

[3]

Patient-specific iPS cells can be induced to differentiate into the affected diseased tissue as this will allow the patient's disease to be modelled *in vitro*. Using this diseased tissue, potential drugs to treat the disease can be screened for, aiding in the discovery of new drugs.

Scientists have managed to obtain iPS cells from cystic fibrosis (CF) patients with a mutation at amino acid position 508 (F508). They have caused these iPS cells to differentiate into diseased human lung tissue, which is now being used to screen for potential drugs to treat cystic fibrosis caused by the F508 mutation (Fig. 1.2).

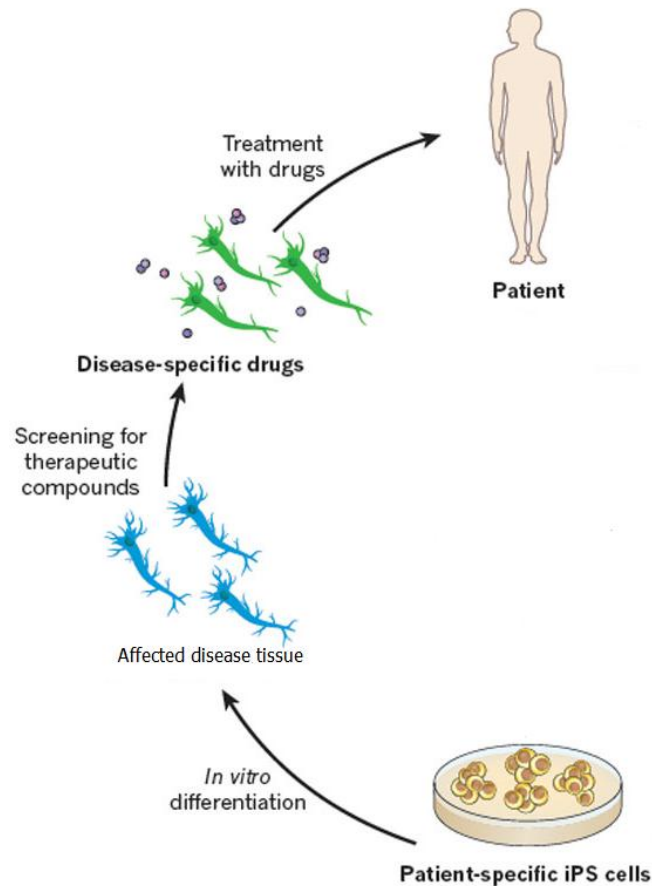


Fig. 1.2

Previously, a cell line comprising rat cells expressing recombinant human mutant CFTR protein was used for screening purposes. This cell line was successfully used to identify a drug to treat cystic fibrosis.

(b) (i) Explain the effect of the F508 mutation on the function of the CFTR protein.

[3]

- (ii) Suggest and explain **one** advantage of using a drug to treat CF rather than to use liposomes to deliver a functional copy of a CFTR gene.

.....
..... [1]

- (iii) Suggest and explain **one** advantage of using diseased lung tissue derived from human iPS cells to screen for CF drugs rather than the use of rat cells expressing recombinant human CFTR.

.....
..... [1]

[Total: 13]

2. Cysteine proteinase (CP) is an enzyme that is coded by the *mir-1* gene, which is found in plants that are naturally resistant to attacks by *Lepidopteran* larvae. *Lepidopteran* larvae feeding on the Black Mexican Sweet (BMS) maize leaf tissue cause crop losses. In an attempt to increase the yield of BMS maize, plant biologists have used genetic technology to transform the wild-type BMS maize callus with *mir-1* gene. The growth of larvae feeding on transformed and non-transformed BMS maize calli was studied for seven days, and the experiment was repeated. The results are shown in Fig. 2.1.

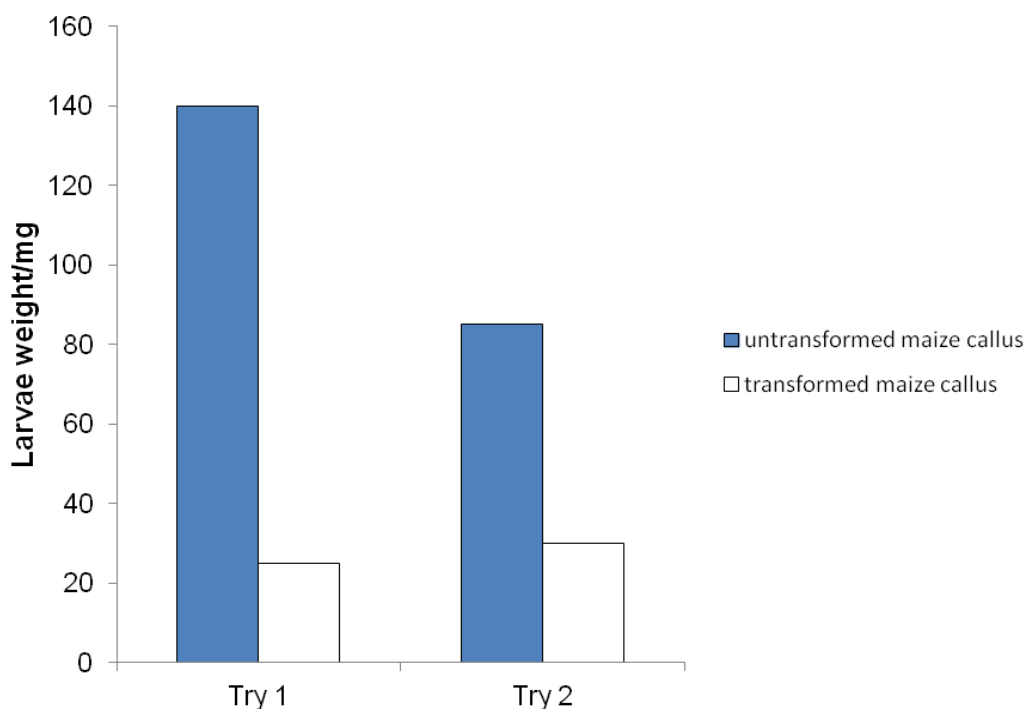


Fig. 2.1

- (a) (i) Suggest **one** delivery method that could be used to introduce the *mir-1* gene into BMS maize callus.

.....
 [1]

- (ii) With reference to Fig. 2.1, describe and explain the effectiveness of genetically engineered BMS maize with *mir-1* gene on larvae growth.

.....

 [3]

- (b) An *in vitro* experimental study was performed to investigate the effects of transformed BMS plant cells expressing both cysteine proteinase (CP) and *Bacillus thuringiensis* (Bt) toxin on *Lepidopteran* larvae. Table 2.1 shows the relative growth rate and percentage mortality of *Lepidopteran* larvae fed with maize cell lines containing CP, Bt-toxin and a combination of CP and Bt-toxin.

Table 2.1

	CP	Bt-toxin	CP+ Bt-toxin
Relative Growth Rate/ Arbitrary Units	0.333	0.568	0.220
Percentage Mortality/%	38.0	4.0	50.0

With reference to Table 2.1, describe the effects of maize plant cells expressing CP together with Bt-toxin on the percentage mortality of *Lepidopteran* larvae.

[2]

- (c) Both CP and Bt-toxin target the mid-gut lumen. Fig. 2.2 shows a cross section of mid-gut. CP is known to damage the peritrophic protein matrix of mid-gut lumen in the larvae, a structure that protects the mid-gut from microbial infection.

The mechanism of damage by Bt-toxin is different compared to that of CP. Bt-toxin is converted to its active form in the alkaline mid-gut environment, which then binds to the specific mid-gut receptors on the brush border membrane of mid-gut cells with very high affinity, forming lytic pores that result in cell death.

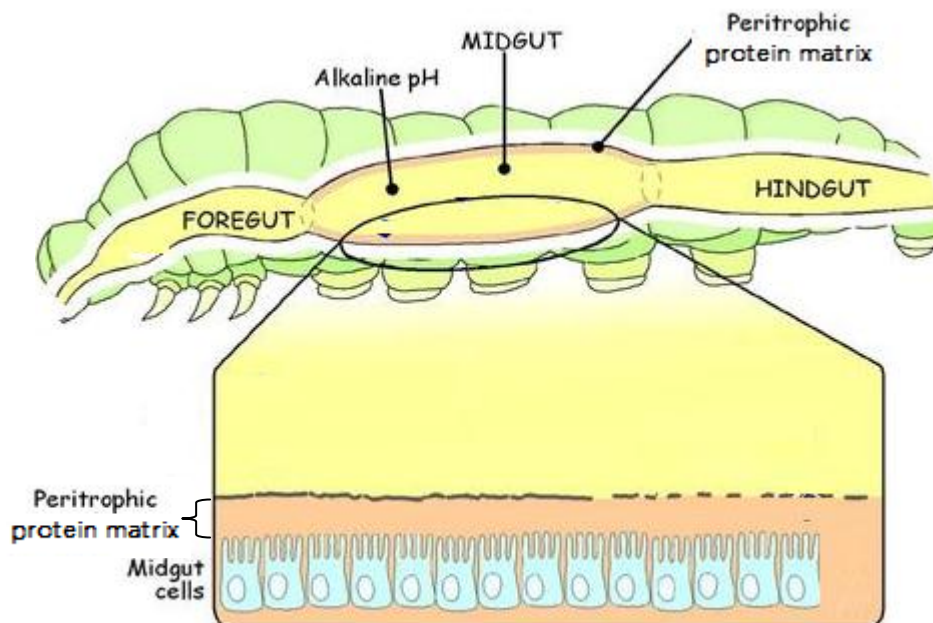


Fig. 2.2

- (i) With reference to Table 2.1, Fig. 2.2 and the information given, suggest how CP may enhance the effect of Bt-toxin on *Lepidopteran* larvae.

[3]

- (ii) Over many generations, laboratory populations of *Lepidopteran* larvae have evolved resistance against Bt-toxins. Suggest and explain how this resistance is developed.

[2]

[Total: 11]

3. Fig. 3.1 below shows a schematic diagram of targeted gene replacement in a mammalian cell involving the gene *Ig M*. Targeted *Ig M* gene replacement is a method of replacing *Ig M* mutated allele (Fig. 3.1a) in the genome in a cell with wildtype allele (Fig. 3.1b) via crossing over, giving rise to the recombinant DNA (Fig. 3.1c). The wildtype allele is carried by a linearised plasmid (Fig. 3.1b) and introduced into the cell.

As shown in Fig. 3.1b, the linearised plasmid contains an antibiotic resistance gene (*Neo gene*), wildtype allele with the exons represented by the black boxes, as well as several restrictions sites in between the exons. In this technique, the enhancer of the *Neo gene* on the linearised plasmid was removed prior to the therapy.

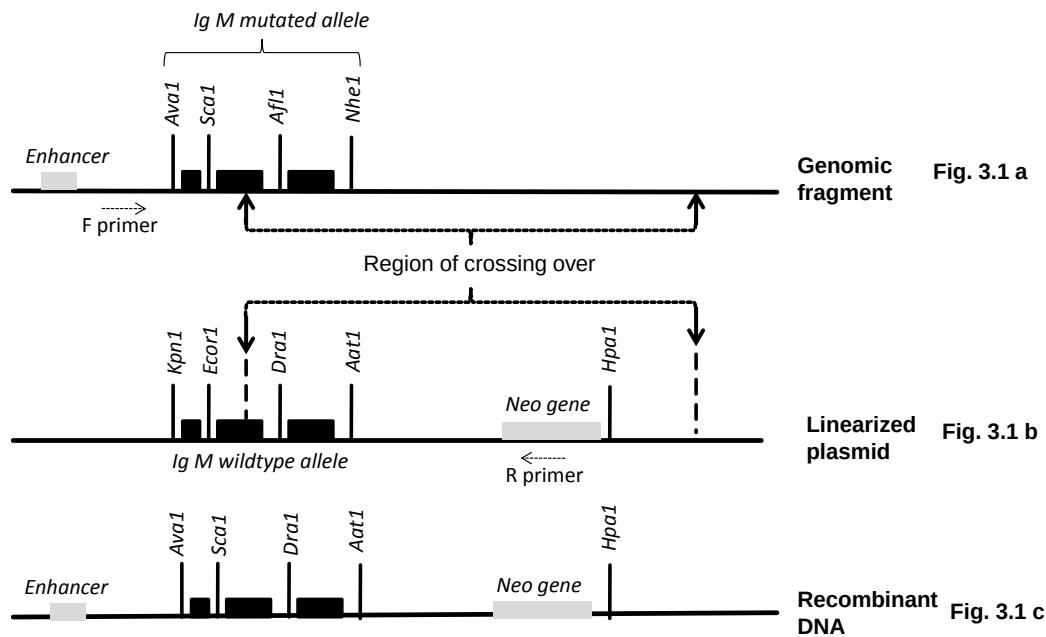


Fig. 3.1

Legend	
Exon	
Restriction sites on mutated allele	<i>Ava1</i> , <i>Sca1</i> , <i>Afl1</i> , <i>Nhe1</i>
Restriction sites on wildtype allele	<i>Kpn1</i> , <i>Ecor1</i> , <i>Dra1</i> , <i>Aat1</i>
Naturally occurring restriction site on linearised plasmid	<i>Hpa1</i>

- (a) (i) Explain the significance of restriction enzyme in bacteria.

[1]

- (ii) Although the mutated and wildtype alleles are homologous regions, different restriction sites are present in them. Explain the presence of these different restriction sites.

[2]

- (iii) With reference to Fig. 3.1, explain how the *Neo* gene allows the transformed cells with recombinant DNA to be distinguished from the transformed cells which did not undergo recombination in their DNA.

[4]

- (iv) With reference to Fig. 3.1a and Fig. 3.1b, briefly describe another method how *Hpa1* can be used to verify the identity of cells with recombinant DNA.

[4]

- (v) The wildtype *IgM* allele was isolated from a genomic DNA library and not a cDNA library before being introduced into the linearised plasmid. With reference to Fig. 3.1, state **one** evidence which suggests this.

[1]

- (b) In the gene replacement therapy shown in Fig. 3.1, the site of DNA breakage occurred between restriction sites *Sca1* and *Afl1* as shown in Fig. 3.1a. In another case of therapy, the site of DNA breakage along the mutated allele was unknown. To determine the exact site of crossing over between the genomic DNA and the linearised plasmid, RFLP analysis was carried out on the recombinant DNA. The recombinant DNA was isolated from the transformed cell and amplified using PCR via the forward and reverse primers as shown in Fig. 3.1a and Fig. 3.1b. The amplified DNA products were separated into eight tubes and digested using eight restriction enzymes respectively: *Ava1*, *Sca1*, *Afl1*, *Nhe1*, *Kpn1*, *Ecor1*, *Dra1*, *Aat1*. The products of digestions were separated using gel electrophoresis and the results are shown in Fig. 3.2.

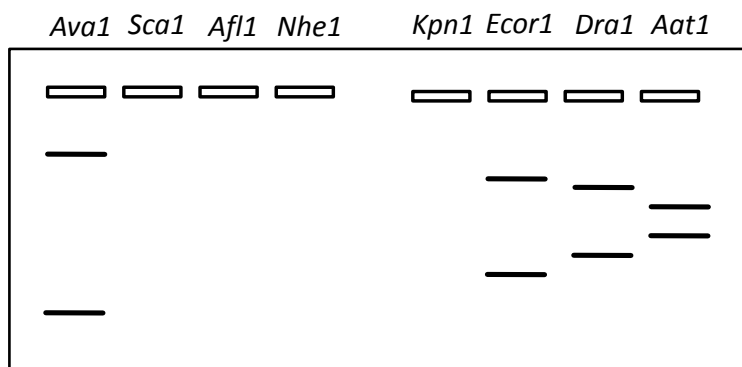


Fig. 3.2

- (i) With reference to Fig. 3.1 and 3.2, describe and explain the band pattern of the restriction digest for the recombinant cell.

[3]

- (ii) Hence identify the site of DNA breakage where crossing over took place in the mutated allele.

[1]

[Total: 16]

Planning Question

4. Electroporation is a method which is routinely used to introduce foreign DNA into *E. coli* cells. This involves subjecting a mixture of bacteria and plasmid to a brief electric pulse. You are required to plan, but not carry out, an investigation into the effect of voltage on the efficiency of transformation by electroporation.

In this investigation, the recombinant plasmid used to transform the *E. coli* cells is made up of two components: the plasmid vector and the gene of interest. pGFP is a plasmid with two marker genes, GFP and *Amp^R*, and a number of restriction sites throughout the plasmid as shown in Fig. 4.1 below. GFP is a marker gene coding for the green fluorescent protein. Expression of GFP in bacteria will cause bacteria to fluoresce under UV light. The gene of interest is the *ras* proto-oncogene, which has been inserted into the plasmid using the restriction enzyme *BmtI*.

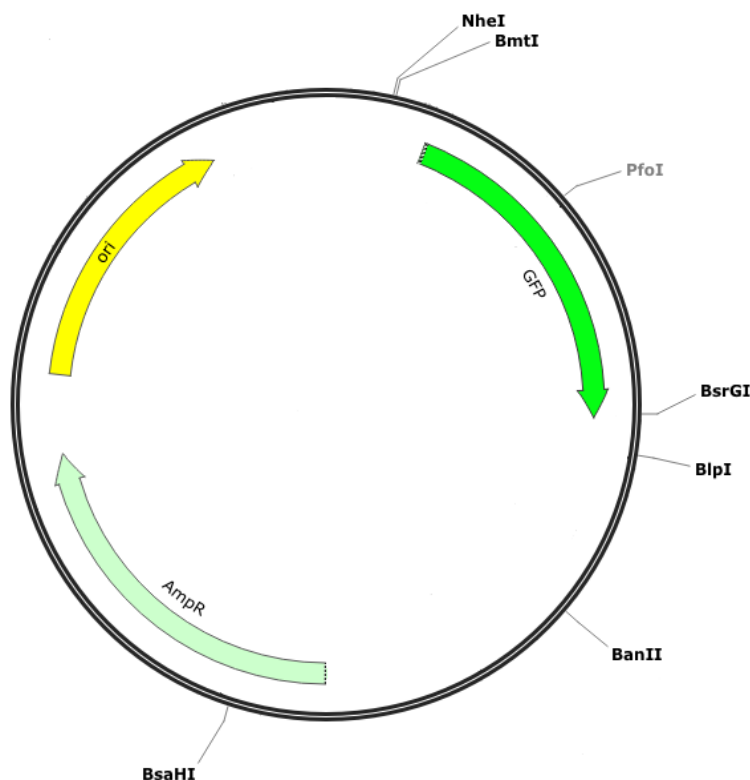


Fig. 4.1

An electroporator is a machine used to carry out electroporation. It is possible to vary the voltage and length of the electrical pulse with this machine. The range of voltage used is typically 1400 – 2000 V and the length of electrical impulse between 4-6 ms. A voltage which is too high will cause irreversible changes in membrane permeability. The mixture of bacteria and recombinant plasmid, which must be kept on ice prior to electroporation, is put into a cuvette (Fig. 4.2) which is then placed into a chamber in the electroporator (Fig. 4.3).

After electroporation, the mixture of bacteria must be quickly resuspended in SOC broth (a type of growth medium) and incubated at 37°C for 1 hour for the surviving cells to recover. This is then followed by plating the bacteria on agar and incubation at 37°C for 16-24 hours. The plates can then be examined to calculate transformation efficiency.

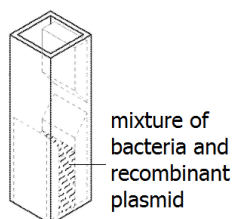


Fig. 4.2

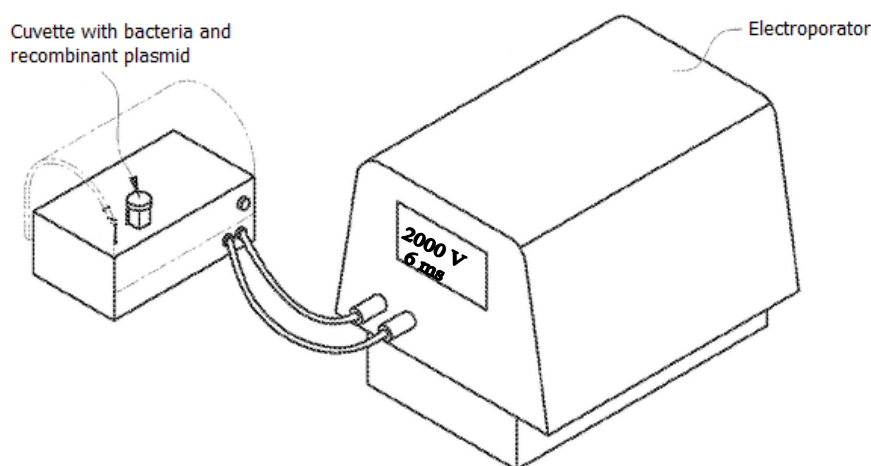


Fig. 4.3

Transformation efficiency can be calculated using the following equation: -

$$\text{Transformation efficiency / cfu } \mu\text{g}^{-1} = \frac{\text{Total number of colonies on agar plate}}{\text{Amount of DNA plated (in } \mu\text{g)}}$$

[cfu: colony forming unit]

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- Microcentrifuge tubes of **100 μl** *E. coli* cells in electroporation buffer
- Microcentrifuge tubes of **1 μl** recombinant DNA (concentration of 0.1 $\mu\text{g} / \mu\text{l}$)
- Microcentrifuge tubes of **1 μl** sterile water
- Ice bucket with ice
- Electroporator
- Sterile cuvettes for electroporation (max. volume of 150 μl)
- Agar plates with ampicillin
- Sterile spreader for spreading bacterial culture on agar plate
- SOC broth
- Incubator
- UV transilluminator (a device which emits UV light)
- UV shields which provide protection from UV light
- A variety of micropipettes, sterile micropipette tips and sterile 2 ml microcentrifuge tubes

Your plan should have a clear and helpful structure to include

- an explanation of the theory to support your practical procedure
- a description of the method used, including the scientific reasoning behind the method, a control experiment and any recommended safety measures
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagram/s showing selection of the recombinant bacteria on plates
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

[Turn over

Free-response question

Write your answers to this question on the separate paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5. **(a)** Describe and explain how bacteria can be used to mass produce human insulin without the use of any DNA libraries. [7]
- (b)** Discuss the advantages and the limitations of the Polymerase Chain Reaction. [5]
- (c)** Discuss the implications of the Human Genome Project, including the benefits and difficult ethical concerns for humans. [8]

[Total: 20]